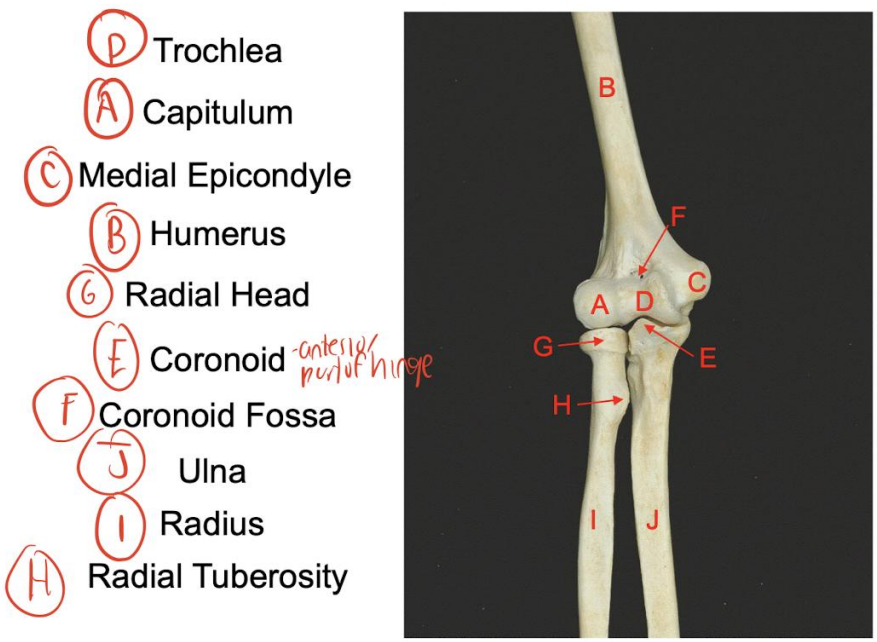
Week #9: [Osteology & Joints of the Distal Upper Extremity]

Anatomy of Bones & Joints

Anatomy of Bones & Joints



I. Osteology:

A. Humerus – only bone of the arm

1. **Radial (spiral) groove** – located on posterior shaft; ***radial nerve*** and ***profunda brachii artery*** run along groove
2. **Medial epicondyle** – medial protuberance at the distal end of the humerus; origin of the ***common flexor tendon*** for the forearm flexors; ***ulnar nerve*** courses posterior to it
3. **Lateral epicondyle** – lateral protuberance at the distal end of the humerus; origin of the ***common extensor tendon*** for the forearm extensors
4. **Capitulum** – lateral condyle; articulates with the head of the radius
5. **Trochlea** – medial condyle; articulates with the *trochlear notch* of the ulna
6. **Coronoid fossa** – depression superior to the trochlea anteriorly; articulates with the coronoid process during full flexion
7. **Olecranon fossa** – depression superior to the trochlea posteriorly; articulates with the olecranon process during full extension

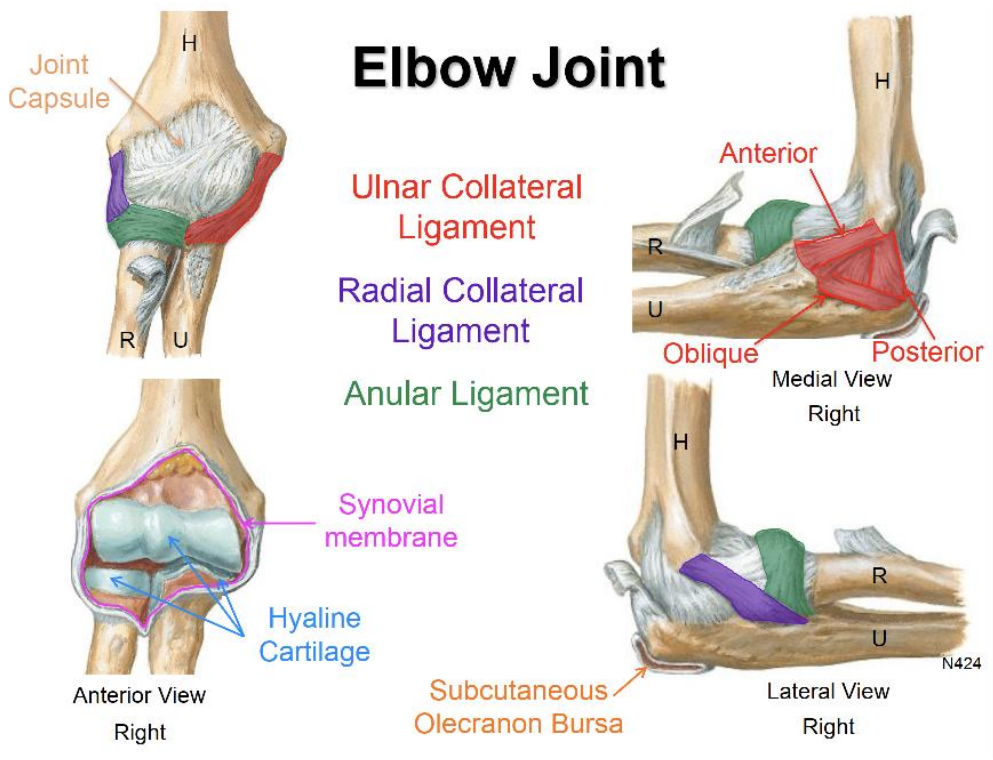
Clinical Note: Fractures of the **surgical neck** of the humerus, often caused by low energy falls to the hand are especially common in elderly individuals with osteoporosis. This type of fracture may result in damage to the ***axillary nerve.*** Transverse fractures of the shaft of the humerus frequently result from a direct blow to the arm. This may result in injury to the ***radial nerve*** and/or ***deep brachial artery.*** Fractures of the distal humerus may damage the ***median nerve.*** Fractures of the **medial epicondyle** may result in damage to the ulnar nerve.

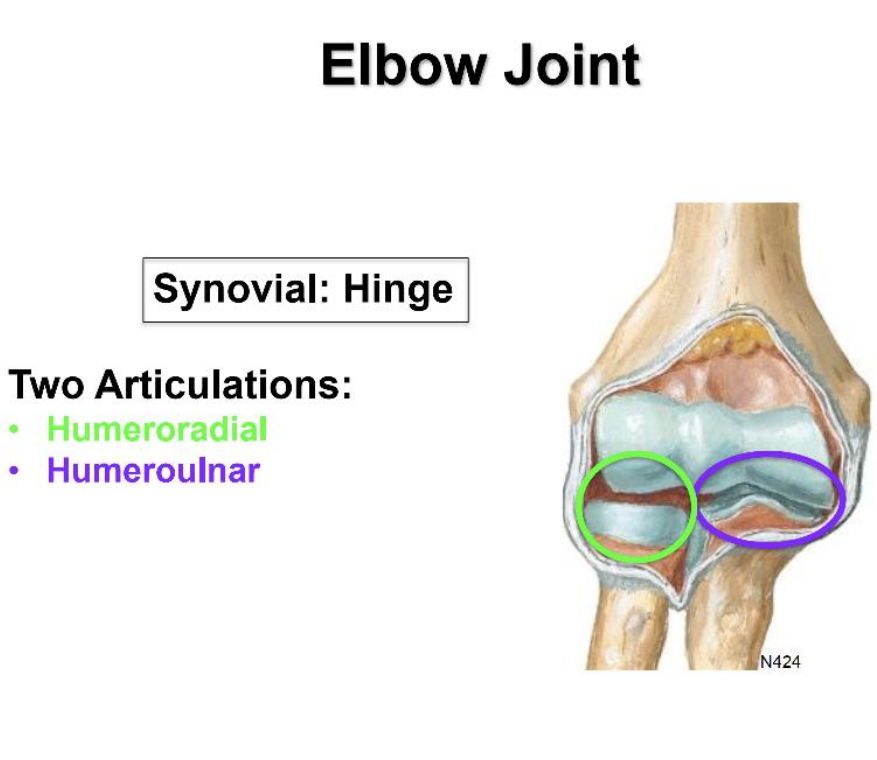
B. Radius – lateral and shorter bone of the forearm

1. **Head** – proximal portion that articulates with the capitulum of the humerus and radial notch of the ulna; pivots within the ***annular ligament***
2. **Neck** – narrow portion between the head and the radial tuberosity
3. **Radial tuberosity**- medial projection; insertion site for ***biceps brachii***

C. Ulna – medial and longer stabilizing bone of the forearm.

1. **Olecranon process** – bony eminence at posterior aspect of proximal ulna; insertion site for ***triceps brachii***
2. **Coronoid process** – triangular protuberance at anterior aspect of proximal ulna; insertion site of ***brachialis***
3. **Trochlear notch** – large depression between olecranon and coronoid process; articulates with trochlea
4. **Radial notch** – concavity on lateral side of coronoid process that articulates with the head of the radius





II. Joints of the Forearm

A. Elbow Joint

1. ***Type:*** hinge type of synovial joint; uniaxial
2. ***Articulation:*** humeroulnar (trochlea of humerus; trochlear notch of ulna) and humeroradial (capitulum of humerus; head of radius)

Note: The elbow joint capsule is weak anteriorly and posteriorly but reinforced medially and laterally by collateral ligaments; muscles help to strengthen the joint anteriorly and posteriorly; the elbow has several prominent bursae (i.e. subcutaneous olecranon bursae, biceps bursa) that help to facilitate movement.

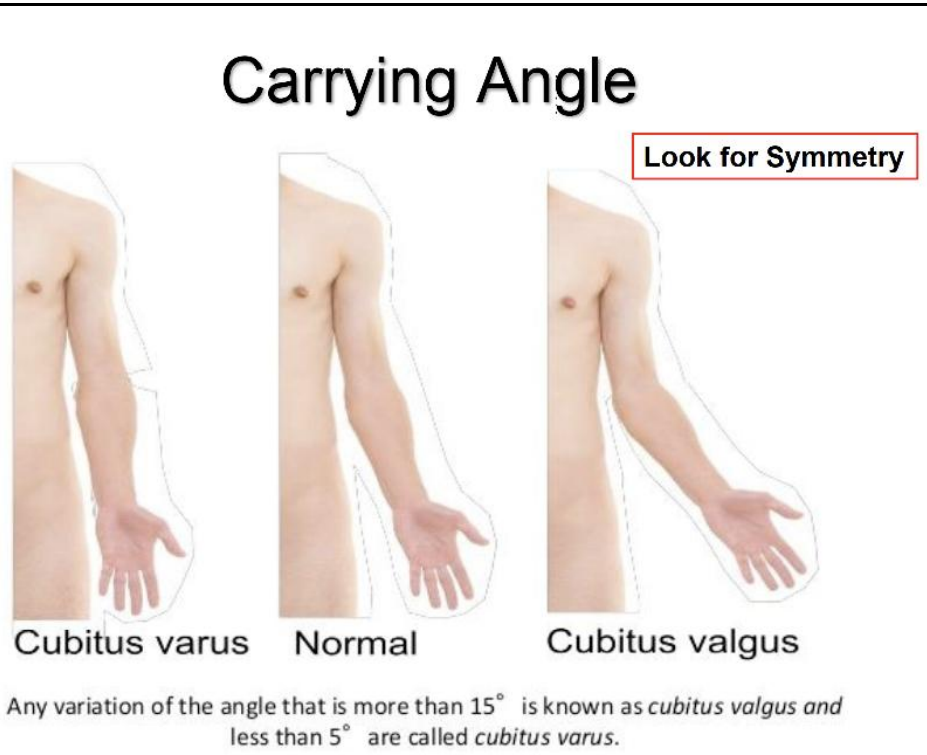
3. ***Ligaments:***
   - ***a.*** Radial collateral ligament: extends from lateral epicondyle of humerus to the head of the radius where fibres blend with annular ligament of radius
   - ***b.*** Ulnar collateral ligament: extends from medial epicondyle of humerus to coronoid process and olecranon of ulna. Consists of three bands: anterior (cord-like, strongest), posterior (fan-like, weakest), and oblique (deepens socket for trochlea of humerus)

Clinical Note: replacement of a damaged ulnar collateral ligament with a muscle tendon (Tommy John surgery) somewhat common in baseball pitchers; 1-2 years of physical therapy required

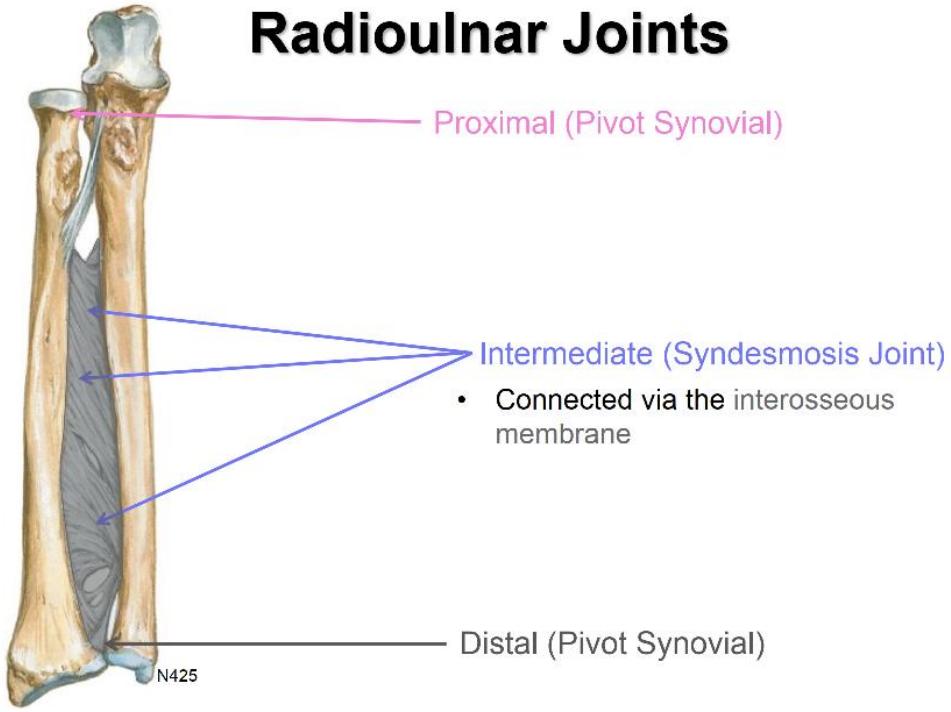
4. ***Movements:*** flexion/extension; review primary muscles involved.
5. ***Blood supply:*** many arteries participate in supplying the elbow joint, including the superior and inferior ulnar collateral a., the anterior and posterior ulnar recurrent a., the radial collateral br. of the deep artery of the arm, and the recurrent radial a.

Clinical Note: posterior dislocation of the elbow joint most common; may be associated with torn joint capsule, ruptured collateral ligaments and bone damage. Ulnar

nerve is at risk due to its position posterior to the medial epicondyle.

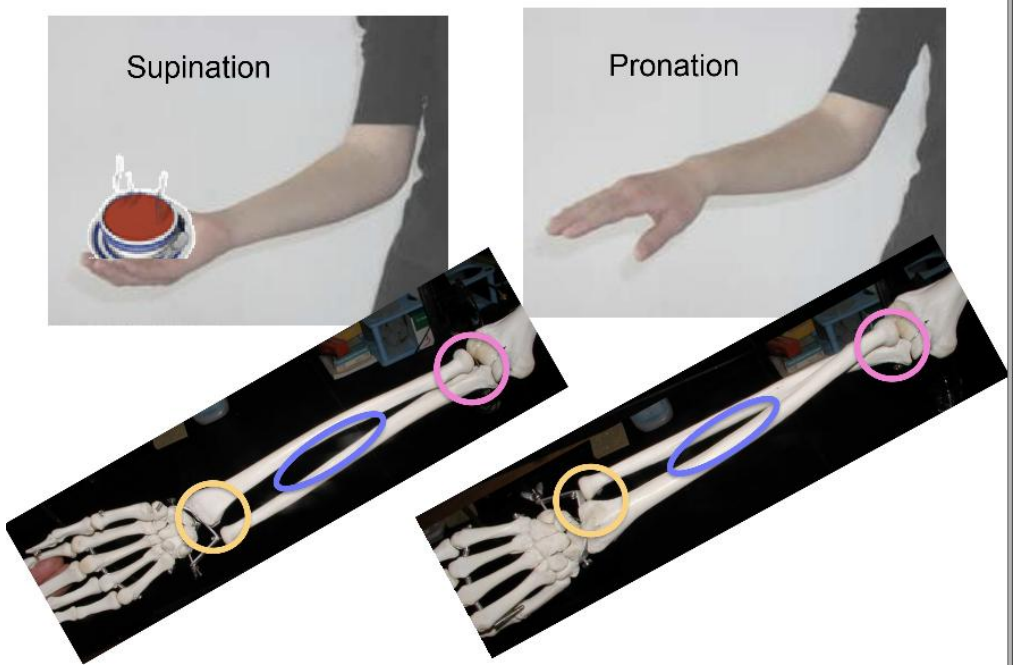


Radioulnar Joints

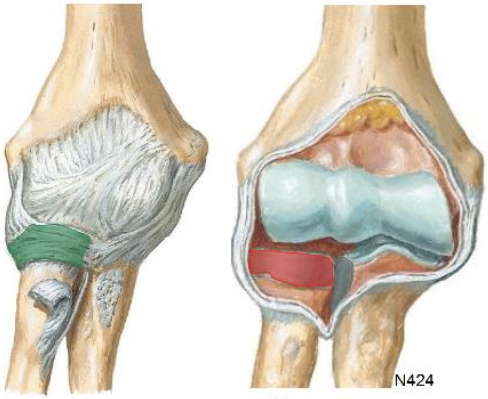


- Connected via the interosseous membrane

Movement of Radioulnar Joints

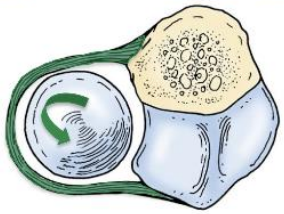


Proximal Radioulnar Joint



Synovial: Pivot Joint

- Articulation of head of radius & radial notch of ulna
- Located within loose fibrous capsule/synovial membrane of elbow joint



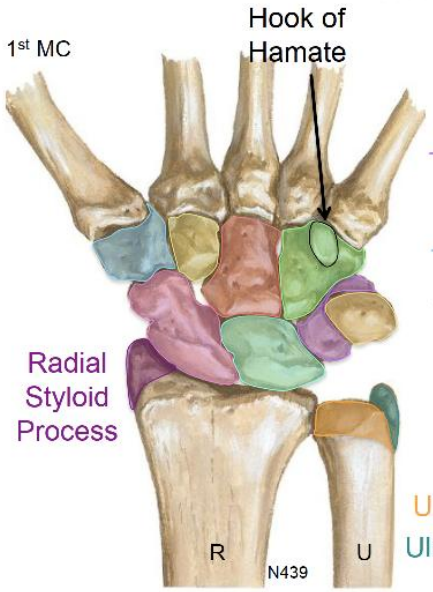
- Anular Ligament allows rotation

B. Proximal Radio-Ulnar Joint

1. ***Type:*** **pivot** type of synovial joint; uniaxial
2. ***Articulation:*** **head of the radius; radial notch of the ulna;** located within fibrous joint capsule of the elbow joint; shares a synovial membrane with the elbow joint
3. ***Ligament:*** the ring-like annular ligament of the radius encircles head of radius
4. ***Movements:*** supination/pronation; head of the radius rotates within annular ligament

Clinical Note: "Nursemaid's Elbow" (subluxation of the proximal radio-ulnar joint); Common in preschool aged children, due to a pulling of the distal aspect of the child's upper limb; typically painful due to pinching of the well-innervated annular ligament and/or joint capsule.

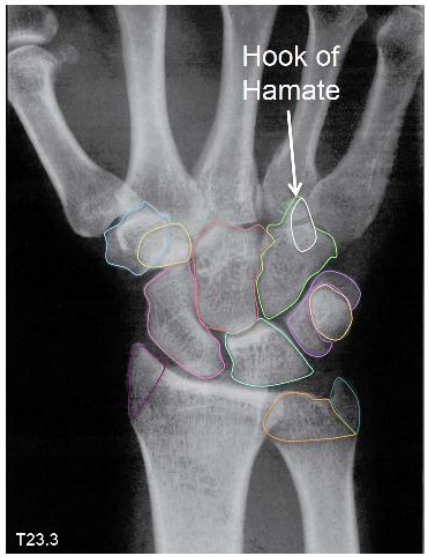
Wrist Osteology



Carpals
- Scaphoid
- Lunate
- Triquetrum
- Pisiform
- Trapezium
- Trapezoid
- Capitate
- Hamate

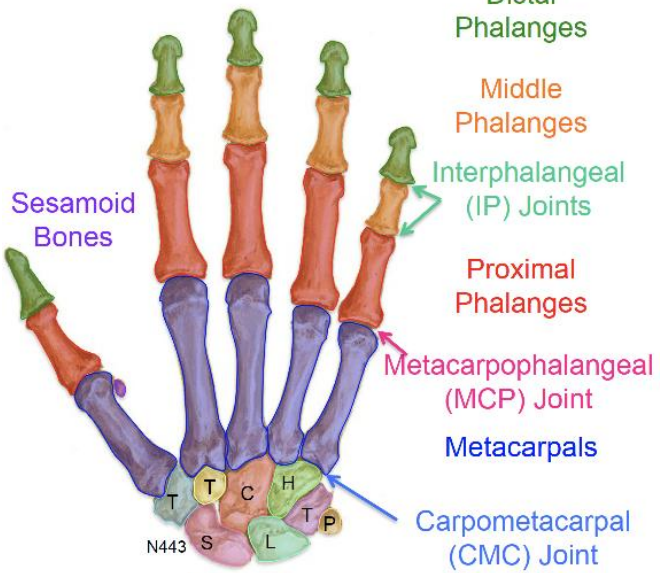
Ulnar Head

Ulnar Styloid Process



Some Lovers Try Positions That They Can't Handle

Hand Osteology



Dorsal View

Distal Phalanges

Middle Phalanges

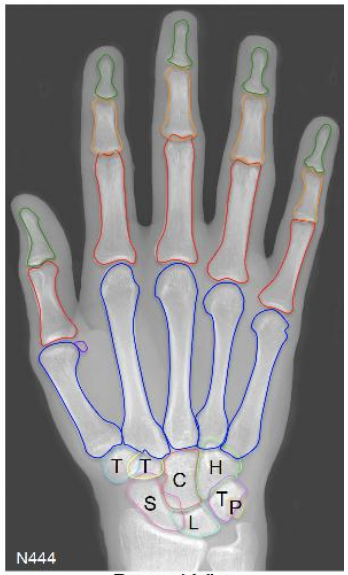
Interphalangeal (IP) Joints

Proximal Phalanges

Metacarpophalangeal (MCP) Joint

Metacarpals

Carpometacarpal (CMC) Joint

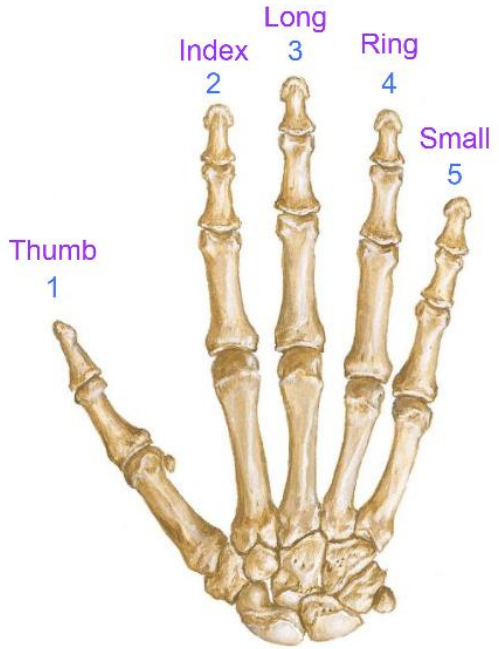


Dorsal View

Digits

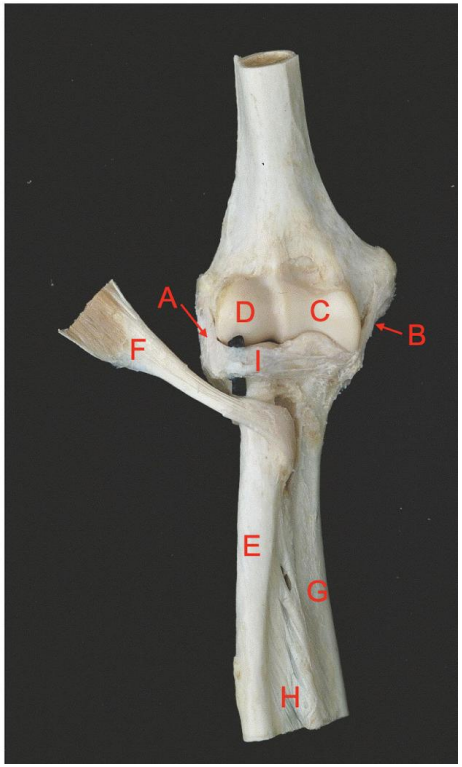
Clinical

Anatomical



Knowledge Check

Bicipital Tendon F
Ulnar Collateral Ligament B
Capitulum D
Ulna G
Trochlea C
Interosseous Membrane H
Radial Collateral Ligament A
Radius E
Anular Ligament I



Knowledge Check

Trapezium F
Triquetrum B
Hamate D
Capitate G
Trapezoid C
Lunate H
Pisiform A
Scaphoid E

Anterior View of Right Hand

